Experiment No. 23 – Top Hat Morphological Operation

1. Aim

To implement the Top Hat technique as a Morphological Operation using OpenCV to extract small bright details from the original image.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Image loading failed")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply Top Hat operation
tophat = cv2.morphologyEx(
    gray, cv2.MORPH_TOPHAT, kernel
)

# Display output
plt.figure(figsize=(12, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(tophat, cmap="gray")
plt.title("Top Hat Image")
plt.axis("off")
```

```
plt.tight_layout()

plt.show()
```

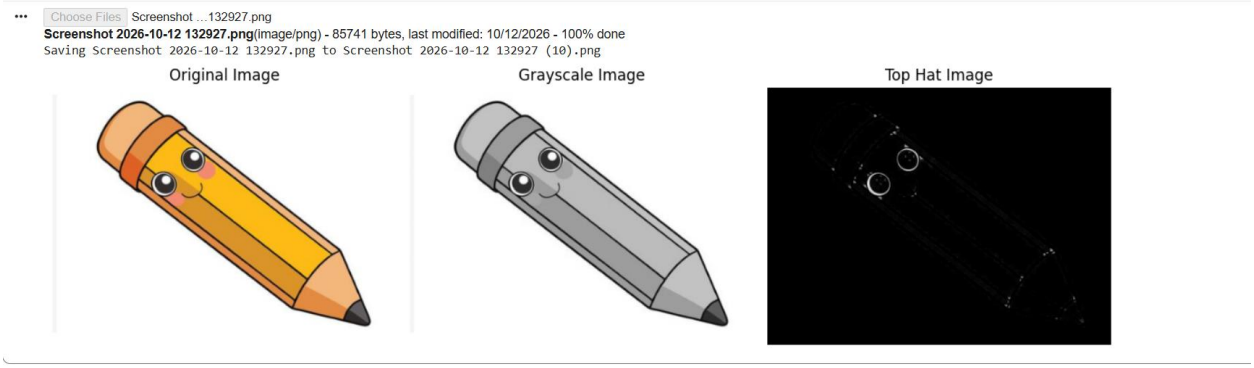
3. Expected Output

The output displays:

- Original Image.
- Grayscale Image.
- Top Hat Image.

The Top Hat operation highlights small bright regions and details that are removed by the opening operation.

4. Output



The Top Hat morphological operation is successfully applied to extract small bright details from the image.

5. Result

Thus, the Top Hat Morphological Operation was successfully implemented using Python and OpenCV.